

CAUSES (INITIATORS)

EFFECTS (FINAL ELEMENTS / ACTIONS) -->

	SIF	INITIATOR	SENSOR(S)		DESCRIPTION	VOTING	NORMAL	TRIP SP	SIL	RESP.	XV- H-101	XV- H-101	XV- H-101	PV- H-101	K-1- H-101	XV- H-201	XV- H-201	FV- H-201	XY- D-101	XY- D-101	XV- C-101	P-1- Crude	P-1- Crude	P-1- Desalt	XV- AR to H-201	P-1- AR / H-201	XV- LPG	XV- C-105	EIV- P-112	EIV- P-204	P-2- V8 & P-204	HV- Heater	F&G Zone	CCR CCR	OSBL Signal					
1		SIF-101	FZLL-1011	FT-1011A/B/C	H-101 pass 1 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																			A						
2		SIF-101	FZLL-1012	FT-1012A/B/C	H-101 pass 2 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
3		SIF-101	FZLL-1013	FT-1013A/B/C	H-101 pass 3 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
4		SIF-101	FZLL-1014	FT-1014A/B/C	H-101 pass 4 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
5		SIF-101	FZLL-1015	FT-1015A/B/C	H-101 pass 5 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
6		SIF-101	FZLL-1016	FT-1016A/B/C	H-101 pass 6 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
7		SIF-101	FZLL-1017	FT-1017A/B/C	H-101 pass 7 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
8		SIF-101	FZLL-1018	FT-1018A/B/C	H-101 pass 8 flow low-low	2oo3	71.1 t/h	28.5 t/h	SIL 2	2 s	X	O	X	P	P																				A					
9		SIF-102	PZLL-1027	PT-1027A/B/C	H-101 burner FG pressure low-low	2oo3	1.0-2.0 barg	0.15 barg	SIL 1	2 s	X	O	X	P	P																				A					
10		SIF-103	PZHH-1027	PT-1027A/B/C	H-101 burner FG pressure high-high	2oo3	1.0-2.0 barg	2.2 barg	SIL 1	2 s	X	O	X	P	P																				A					
11		SIF-104	BZLL-1028	BS-1028 (16)	H-101 loss of flame (all burners)	per burner / all	flame on	flame off	SIL 2	4 s (FFRT)	X	O	X	P	P																				A					
12		SIF-105	PZHH-1029	PT-1029A/B/C	H-101 arch pressure high-high / ID fan loss	2oo3	-2.5 mmH2O	+2.5 mmH2O	SIL 1	3 s	X	O	X	P	P																					A				
13		SIF-106	TZHH-1020	TT-1020A/B/C	H-101 COT high-high	2oo3	366 °C	381 °C	SIL 1	5 s	X	O	X	P	P																					A				
14		SIF-107	LZLL-1007	LT-1007B	D-101A interface low-low (grid short)	1oo1	50 %	15 %	SIL 1	5 s										X																	A			
15		SIF-107	LZLL-1008	LT-1008B	D-101B interface low-low (grid short)	1oo1	50 %	15 %	SIL 1	5 s										X																	A			
16		SIF-108	LZHH-1082	LT-1082B	C-101 bottom level high-high	1oo1	50 %	85 %	SIL 1	10 s												X	X														A			
17		SIF-109	LZLL-1092	LT-1092B	D-105 level low-low (gas blow-by to LPG)	1oo1	50 %	15 %	SIL 1	5 s																		X									A			
18		SIF-110	PZHH-1091	PT-1091B	C-105 pressure high-high	1oo1	10.8 barg	12.2 barg	SIL 1	5 s																			X								A			
19		SIF-201	FZLL-2001	FT-2001A/B/C	H-201 pass 1 flow low-low	2oo3	64.5 t/h	25.8 t/h	SIL 2	2 s						X	O	O																		A				
20		SIF-201	FZLL-2002	FT-2002A/B/C	H-201 pass 2 flow low-low	2oo3	64.5 t/h	25.8 t/h	SIL 2	2 s						X	O	O																			A			
21		SIF-201	FZLL-2003	FT-2003A/B/C	H-201 pass 3 flow low-low	2oo3	64.5 t/h	25.8 t/h	SIL 2	2 s						X	O	O																			A			
22		SIF-201	FZLL-2004	FT-2004A/B/C	H-201 pass 4 flow low-low	2oo3	64.5 t/h	25.8 t/h	SIL 2	2 s						X	O	O																			A			
23		SIF-202	PZLL-2009	PT-2009A/B/C	H-201 burner FG pressure low-low	2oo3	1.0-2.0 barg	0.15 barg	SIL 1	2 s						X	O	O																			A			
24		SIF-202	BZLL-2008	BS-2008 (6)	H-201 loss of flame (all burners)	per burner / all	flame on	flame off	SIL 1	4 s (FFRT)						X	O	O																			A			
25		SIF-203	LZHH-2024	LT-2024B	C-201 bottom level high-high	1oo1	50 %	85 %	SIL 1	10 s																X	X											A		
26		SIF-204	HS-1121	HS-1121 / BY-1121	P-112 area fire - manual / F&G confirmed	1oo2	-	-	SIL 1	30 s (valve)																X				X								A		
27		SIF-204	HS-2041	HS-2041 / BY-2041	P-204 area fire - manual / F&G confirmed	1oo2	-	-	SIL 1	30 s (valve)																					X	X						A		
28		SIF-901	HS-9000	HS-9000 (CCR + field)	Unit ESD-1 push-button	1oo2	-	-	SIL 2	2 s	X	O	X	P	P	X	O	O	X	X	X	X	X	X	X	X	X	X	X							A	A			
29	F&G	GZHH-01	GD-01xx		Confirmed gas - Desalting & crude charge	2ooN	0	20 / 50 % LEL	F&G	10 s																											A	A		
30	F&G	BZHH-01	BD-01xx / HS-01xx		Confirmed fire - Desalting & crude charge	2ooN / MCP	-	flame	F&G	10 s																												X		
31	F&G	GZHH-02	GD-02xx		Confirmed gas - Preheat exchanger bank	2ooN	0	20 / 50 % LEL	F&G	10 s																														
32	F&G	BZHH-02	BD-02xx / HS-02xx		Confirmed fire - Preheat exchanger bank	2ooN / MCP	-	flame	F&G	10 s																												X		
33	F&G	GZHH-03	GD-03xx		Confirmed gas - C-101 / strippers / OH system & pumps	2ooN	0	20 / 50 % LEL	F&G	10 s																														
34	F&G	BZHH-03	BD-03xx / HS-03xx		Confirmed fire - C-101 / strippers / OH system & pumps	2ooN / MCP	-	flame	F&G	10 s																				X								X		
35	F&G	GZHH-04	GD-04xx		Confirmed gas - Light ends C-105/C-106 & LPG pumps	2ooN	0	20 / 50 % LEL	F&G	10 s																														
36	F&G	BZHH-04	BD-04xx / HS-04xx		Confirmed fire - Light ends C-105/C-106 & LPG pumps	2ooN / MCP	-	flame	F&G	10 s																												X		
37	F&G	GZHH-05	GD-05xx		Confirmed gas - Fired heaters H-101/H-201 & FG KO	2ooN	0	20 / 50 % LEL	F&G	10 s	X		X				X																							
38	F&G	BZHH-05	BD-05xx / HS-05xx		Confirmed fire - Fired heaters H-101/H-201 & FG KO	2ooN / MCP	-	flame	F&G	10 s	X		X				X																O					X		
39	F&G	GZHH-06	GD-06xx		Confirmed gas - VDU C-201, ejectors & hotwell	2ooN	0	20 / 50 % LEL	F&G	10 s																														
40	F&G	BZHH-06	BD-06xx / HS-06xx		Confirmed fire - VDU C-201, ejectors & hotwell	2ooN / MCP	-	flame	F&G	10 s																					X	X							X	

LEGEND: X = trip / close (de-energise) O = open P = purge / safe position, controller to manual-track A = alarm. Shaded rows = F&G confirmed events.

1. SIS / BMS hosted in the SIL 3-capable logic solver (separate from DCS); BMS per NFPA 85/86 & API 556. All trips latch; reset from CCR after initiator healthy and purge / permissives complete.

2. Set points derived from process_results / psv data (set-point basis in RPT-002); vendor confirmation required. F&G: 2ooN confirmed gas (20 % LEL alarm, 50 % LEL action) or fire; layout per mapping study.

3. SIF-204 ROSOVs fire-safe, close <= 30 s. SIF-109 = D-105 LL -> XV-1093 (SIF list revision).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
CAUSE & EFFECT MATRIX			
SIS / BMS / ESD / F&G - CDU / VDU			
DISCIPLINE		STAGE	SCALE
INSTRUMENTATION		FEED	NTS
		SHEET	
		1 OF 1	
DRAWN		CHECKED	APPROVED
CLAUDE CODE		-	-
		DATE	
		2026-10-02	
DRAWING NUMBER			REV
CFU-000-IC-CE-001			A